Harsh Upadhayay

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Professional Summary

Backend & Cloud Engineer with 2+ years of production experience designing distributed systems, event-driven architectures, and AWS cloud-native infrastructure. Owns greenfield architecture and production operations across scraping platforms, observability systems, and enterprise data pipelines; delivered 95%+ runtime reductions, reduced MTTD from 30 minutes to 1 minute, cut AWS spend by $1,000+/month, and built fault-tolerant backend services using ECS Fargate, Lambda, Kafka/RabbitMQ, Celery, and CDK. .

Certifications

- **AWS DevOps Professional (86%)**
- **AWS SysOps Administrator Assoc. (88%)**
- **AWS Developer Associate (86%)**
- **AWS Solutions Architect Professional (86%)**
- **AWS Solutions Architect Assoc. (90%)**

Experience

SMS DataTech — **July 2023 – Present**
Backend & Cloud Engineer — *Tokyo, Japan*

- **AI Scraping Platform – Greenfield System Design:** Owned end-to-end architecture for a distributed, AI-powered scraping suite; designed a fault-tolerant, event-driven orchestration layer (Django/Celery, RabbitMQ) with parallel task batching, automated retries, and Dead Letter Queues. Deployed containerized microservices to ECS Fargate via AWS CDK (IaC), enforcing VPC isolation, Secrets Manager, and Fargate Spot with task idempotency; built custom Lambda/EventBridge auto-scaling to eliminate idle compute waste.
- **Cloud Cost Optimization & Infrastructure Ownership:** Reduced total AWS spend by $1,000+/month as primary Cloud Administrator; cut compute costs $500/month via Savings Plans, Graviton migration, and Spot Instances. Diagnosed a 1.2 TB Aurora RDS bloat, executed a non-blocking batched deletion over 3 days (zero downtime), shrinking the table 83% and cutting RDS costs 50% ($500/month); archived snapshots to Glacier Deep Archive for long-term retention.
- **Enterprise Data Pipeline (Sony Group):** Assumed full ownership of a legacy large-scale collection system across 14 media/e-commerce platforms; refactored 6 major scrapers, migrated normalization logic into Scrapy pipelines, and eliminated intermediary Django/Celery/DB steps – achieving a **95% runtime reduction**. Onboarded and mentored a replacement FTE while maintaining uninterrupted daily/weekly S3 data delivery.
- **Production Data Pipeline (Amazon):** Managed a major mid-project specification pivot, migrating live pipelines to new source sites without service disruption; developed new Scrapy spiders and automated previously manual extraction workflows to reduce operational overhead.
- **PoGo Observability Platform – Event-Driven Monitoring:** Re-architected a polling-based monitor for 20+ servers and 40+ distributed scrapers into an Apache Kafka async pipeline, reducing MTTD from 1 day to 1 min and eliminating all client-reported escaped defects. Built a CPU/RAM/Disk resource dashboard with 14-day history, cutting per-incident monitoring overhead from 3 hours to 5 minutes.

Technical Skills

- **Languages & Frameworks:** Python, C++, C, SQL, Django, FastAPI, Scrapy, Pandas, NumPy, REST APIs, PostgreSQL, MySQL | **Backend:** Distributed Systems, Microservices, Event-Driven Architecture, System Design, Scalability, Fault Tolerance, Async Processing (Celery)
- **Cloud & DevOps:** AWS (ECS Fargate, EC2, Lambda, EventBridge, CloudFormation, CDK, CloudFront, RDS/Aurora, S3, IAM, Secrets Manager, CloudWatch), Docker, Kafka, RabbitMQ, Redis, CI/CD, Linux, IaC | **Observability:** Monitoring, Logging, Alerting, Incident Response, Production Reliability

Projects

Self-Hosted Homelab (40+ Services, AI + Cloud Stack) | *Docker, Traefik v3, Authelia, LLDAP, Prometheus, Grafana, Jenkins, Ollama, Cloudflare*

- Architected a domain-driven, containerized infrastructure with Traefik v3 as edge router, Authelia ForwardAuth + OIDC, and LLDAP for identity management across 40+ services; enforced per-service Zero-Trust access policies with secure Cloudflare ingress and segmented network architecture. Integrated full observability (Prometheus, Grafana), CI/CD pipelines (Jenkins + DinD), and GPU-enabled local AI services (Ollama with OpenAI-compatible API) with centralized stateful storage.

CipherVault | *C++, Cryptography, OOP* — Modular cryptographic library implementing classical cipher algorithms from scratch using object-oriented design principles.

Education

Indian Institute of Information Technology, Nagpur — **May 2024**
Bachelor of Technology in Computer Science and Engineering ***(8.5 CGPA)*** — *Nagpur, India*

Kendriya Vidyalaya O.N.G.C. — **March 2019**
Senior Secondary ***(94.4%)****, High School* ***(10 CGPA)*** — *Dehradun, India*

Achievements & Profiles

- Solved **1000+** DSA problems on **LeetCode** and **Codeforces**.
- Achieved Specialist rating (**1415 max**) on **Codeforces**.
- Published **5+** technical articles on **GeeksforGeeks**.
- Coursera Certs: **Supervised ML** & **Adv. Learning Algorithms**.